

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester II Examination, 2022-23

Paper No. MTB-201: Calculus-III

Time: 3 Hours

Full Marks: 70

Ref. No.: 063898

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Determine the simultaneous limit and repeated limits of $f(x, y) = \frac{x^3 - y^3}{x^2 + y^2}$ at $(0, 0)$.
- (b) By using the definition of limit, show that $\lim_{(x, y) \rightarrow (1, 2)} (x + y^2) = 3$.
- (c) If $z = x^{4/3}y^{2/3}$, where $x = t^3$ and $y = t^2$, find $\frac{dz}{dt}$.
- (d) Investigate for maximum and minimum values of $y = \sin x + \cos 2x$.
- (e) If $f(x, y) = \frac{xy}{\sqrt{x^2 + y^2}}$ for $x^2 + y^2 \neq 0$, find $f_x(0, 0)$ and $f_y(0, 0)$.
- (f) Verify Euler's theorem for $u = x^2 \log\left(\frac{y}{x}\right)$.
- (g) Show that $B(n, n) = \frac{\sqrt{\pi} \Gamma(n)}{2^{2n-1} \Gamma\left(n + \frac{1}{2}\right)}$.
- (h) If $u^a v^b w^c = 1$, show that $\frac{\partial^2 w}{\partial u \partial v} = \frac{-1}{(w \log ew)^3}$ when $u = v = w$.
- (i) Evaluate $\int_0^1 x^4(1-x)^3 dx$.
- (j) Show that $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \cdot \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$.
- (k) Evaluate $\int_0^\infty e^{-x^2} dx$.

Question 2.

- (a) State and prove a necessary and sufficient condition for the existence of limit of $f(x, y)$ at (a, b) . Evaluate $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}$. [6]

- (b) If $f(x, y)$ possesses partial derivatives at every point of a neighbourhood of (a, b) and these are continuous at (a, b) , show that f is differentiable at (a, b) . [6]
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Question 3.

- (a) If $u = f(r)$ where $r^2 = x^2 + y^2$, prove $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$. [6]
- (b) Find the extreme values of $f(x, y) = x^3 + y^3 - 3xy$. [6]
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Question 4.

- (a) Change the order of integration and evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} f(x, y) dy dx$. [6]
- (b) Evaluate $\iint_R \frac{dA}{(1+x^2+y^2)^2}$ where R is the region $x^2 + y^2 \leq 1$. [6]
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Question 5.

- (a) Prove $\Gamma(n+1) = n\Gamma(n)$ and hence $\Gamma(n+1) = n!$ for positive integers. [6]
- (b) Evaluate $\int_0^{\pi/2} \sin^5 \theta \cos^6 \theta d\theta$ using Beta and Gamma functions. [6]
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Question 6.

- (a) State and prove Dirichlet's theorem on multiple integrals. Evaluate $\iiint x^l y^m z^n dx dy dz$ over the region $x + y + z \leq 1, x, y, z \geq 0$. [6]
- (b) Find the volume bounded by $z = x^2 + y^2$ and $z = 2y$. [6]
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Question 7.

- (a) Evaluate $\int_0^\infty \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin(p\pi)}$ for $0 < p < 1$. [6]
- (b) Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xyz dz dy dx$. [6]
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